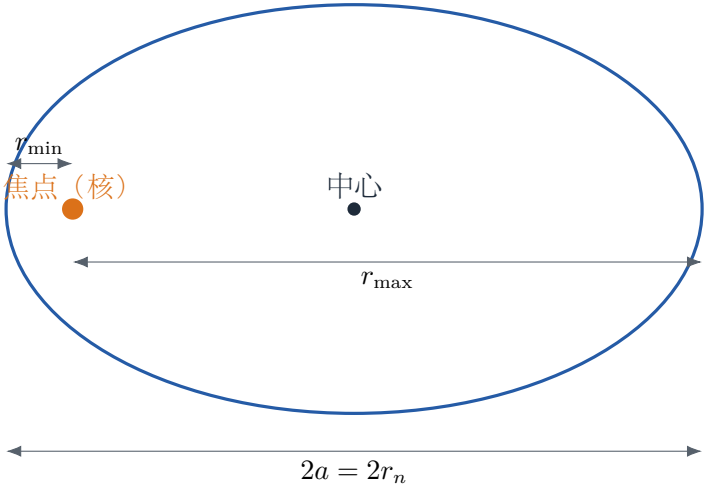

$$a = r_n = \frac{n^2\hbar^2}{m_e\kappa}$$

$$e = \sqrt{1 - \frac{l^2}{n^2}}$$

$$r_{\min} = a(1 - e)$$

$$r_{\max} = a(1 + e)$$